FINAL REPORT

DATASET 3 – NORTHWIND TRADERS

Focus: To analyses sales performance, customer behaviour, product demand, employee productivity, and shipping costs using Power BI to support better business decisions.

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01.INTRODUCTION:

The Northwind Traders Dataset is a sample business dataset used to analyse and understand the operations of a trading company. It contains information about customers, employees, products, product categories, orders, order details, and shipping companies. This dataset helps analyse sales performance, customer purchasing patterns, product demand, and order management.

By combining and analysing this dataset, we can identify key business trends, evaluate sales performance, understand customer behaviour, and support effective decision-making. This report uses the Northwind Traders dataset to gain meaningful insights into business operations and improve overall business performance.

1.1.Purpose of Dashboards

- Analyse overall sales performance.
- Identify top-selling products and categories.
- Understand customer purchasing behaviour.
- Track orders and employee performance.
- Monitor shipping activities.

1.2.Purpose of Business Insights Report

- Identify key business trends and patterns.
- Evaluate sales and product performance.
- Provide recommendations for improvement.
- Support better business decision-making.
- Discover business opportunities.

02.DASHBOARD OVERVIEW:

The following dashboard provides a visual representation of Northwind Traders business performance, highlighting key metrics, sales trends, orders details, product performance, and order analysis.



2.1. Purpose of the Dashboard

To provide an interactive view of Northwind Traders sales and business performance by analysing orders, product categories, employees, and key business metrics.

2.2. Key Components

- Orders by Month – Monthly order trends.
- Orders by Employee – Employee-wise order analysis.
- Sales by Product Category – Category-wise sales performance.
- KPI Cards – Total Orders, Total Quantity, Total Freight, Average Freight, and Number of Customers.

2.3. Interactive Filters Available

- Year: Filter data based on the selected year.
- Region: View business performance by region.
- Category: Analyse sales based on product categories.

03. INTERPRET KPI CARDS:



KPI	VALUE	INTERPRETATION
TOTAL ORDERS	809	Shows 809 orders recorded, helping compare order growth over years.
TOTAL QUANTITY SOLD	51 K	Indicates approximately 51,000 units sold, reflecting changes in product demand over the years.
TOTAL FREIGHT	63.96 K	Represents total shipping expenses, which vary depending on yearly order and delivery volumes.
AVERAGE FREIGHT	79.05	Shows the average freight cost per order, helping identify changes in shipping costs across years.
NUMBER OF CUSTOMERS	91	Represents 91 customers, helping analyse customer growth and retention over the years.

04. PRODUCT CATEGORY PERFORMANCE ANALYSIS:

SALES BY PRODUCT CATEGORY

Highest Selling Category: Beverages

Lowest Selling Category: Grains & Cereals

Relative Differences:

- Beverages generate the highest sales compared to other categories.
- Grains & Cereals contribute the lowest sales.

Observations:

- The Observation made by Clustered Column Chart.
- Customers show higher demand for Beverages.
- Grains & Cereals have comparatively lower sales performance.

Evidence:



Key Insights:

Focus on promoting high-performing categories. Improve sales of low-performing categories through offers and marketing strategies.

05. MONTHLY ORDER TREND ANALYSIS:

ORDERS BY MONTH

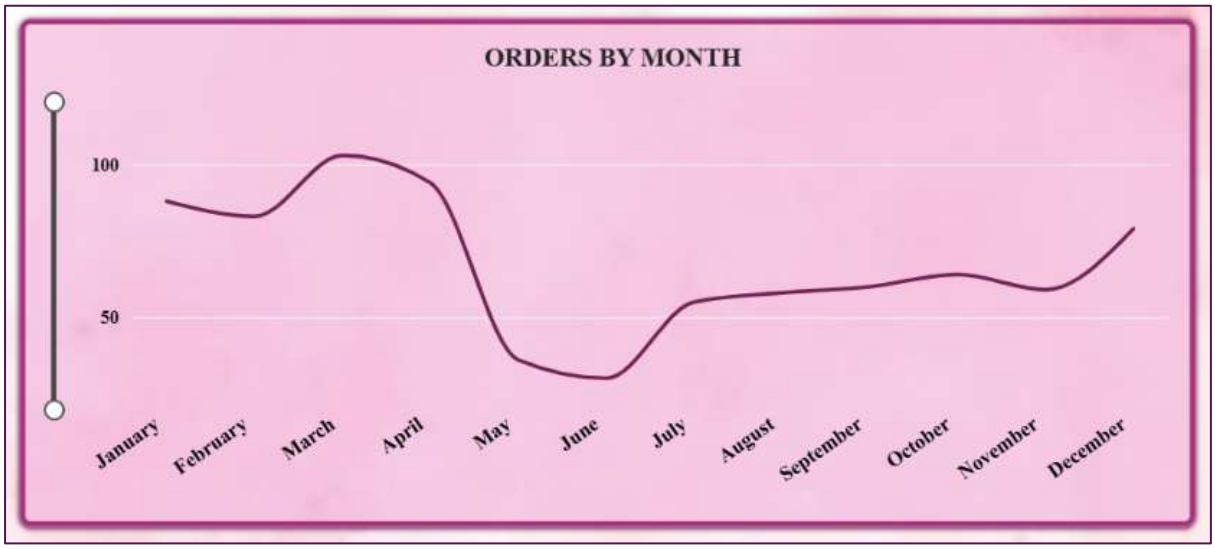
Highest Order Month: March recorded the highest number of orders, with 103 orders.

Lowest Order Month: June recorded the lowest number of orders, with 30 orders.

Overall Pattern:

- The Trend is analysed by using Line Chart.
- Orders fluctuate throughout the months, showing variations in customer demand.

Evidence:



Major Fluctuations:

A significant difference of 73 orders is observed between March and June.

Key Insights:

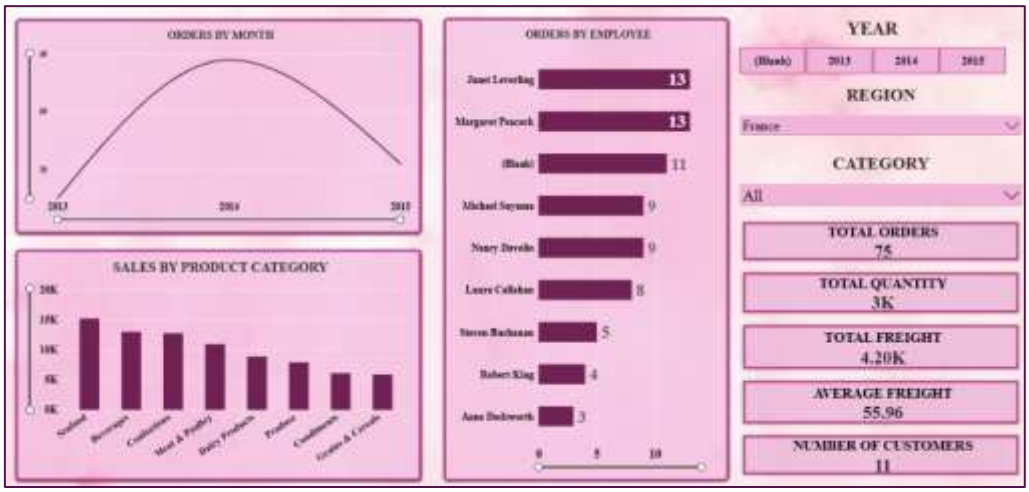
The company should maintain strong order performance during peak months and introduce promotional offers during low-order months to improve sales consistency.

06. REGIONAL / FILTER BASED PERFORMANCE ANALYSIS:

REGION SLICER

REGION	MAJOR OBSERVATION	BUSINESS INSIGHTS
Switzerland	**Dairy products** show a different sales pattern. Orders increased **from 3 in 2013 to 6 in 2015**, with only **2 customers**.	Shows moderate order growth despite having only 2 customers.
Norway	**Beverages** are the best-selling product category. Orders increased **from 1 in 2013 to 3 in 2015**, with **only 1 customer**.	Has a limited customer base but shows growth in orders.
France	Recorded the highest sales of **Seafood**. Orders increased **from 15 in 2013 to 21 in 2015**, with **11 customers**.	Shows strong sales performance, with more customers and increasing orders.

Overview: France is the strongest-performing region, while Norway and Switzerland have opportunities to increase customer numbers and sales through better product-focused marketing strategies.



07. EMPLOYEE PERFORMANCE ANALYSIS:

ORDERS BY EMPLOYEE

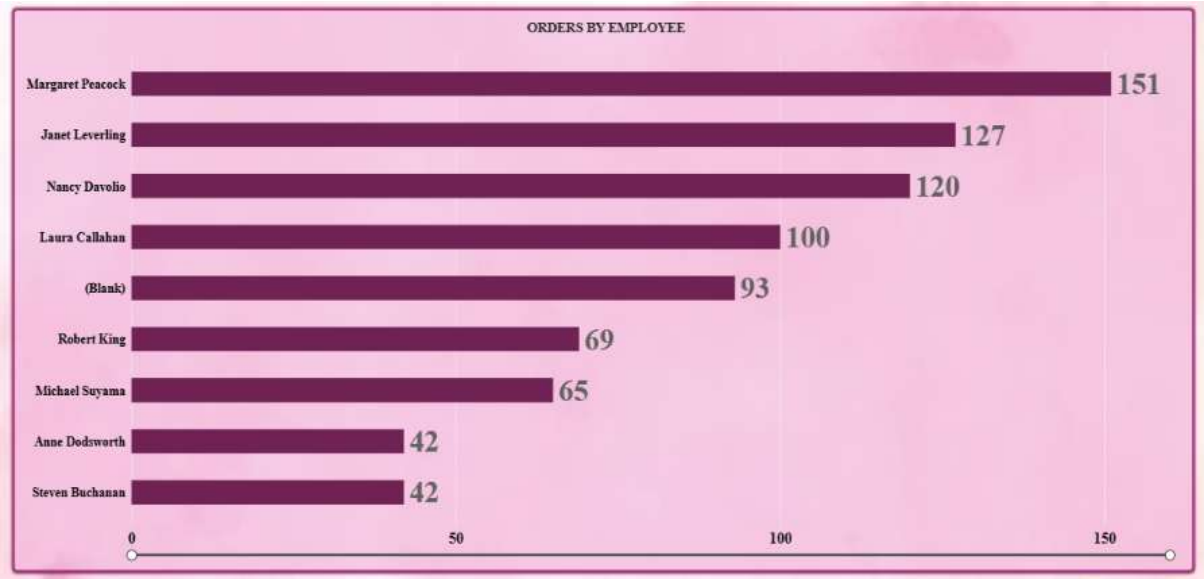
Top-Performing Employee: Margaret Peacock – 151 orders.

Lowest-Performing Employee: Steven Buchanan – 42 orders.

Difference between employees:

114 orders difference between the highest and lowest performers.

Evidence:



Changes after applying filters:

- Margaret Peacock ranked highest in all three years (2013–2015).
- Steven Buchanan recorded the lowest overall orders (42).
- Anne Dodsworth ranked lowest in 2013, while Steven ranked lowest in 2014 and 2015.

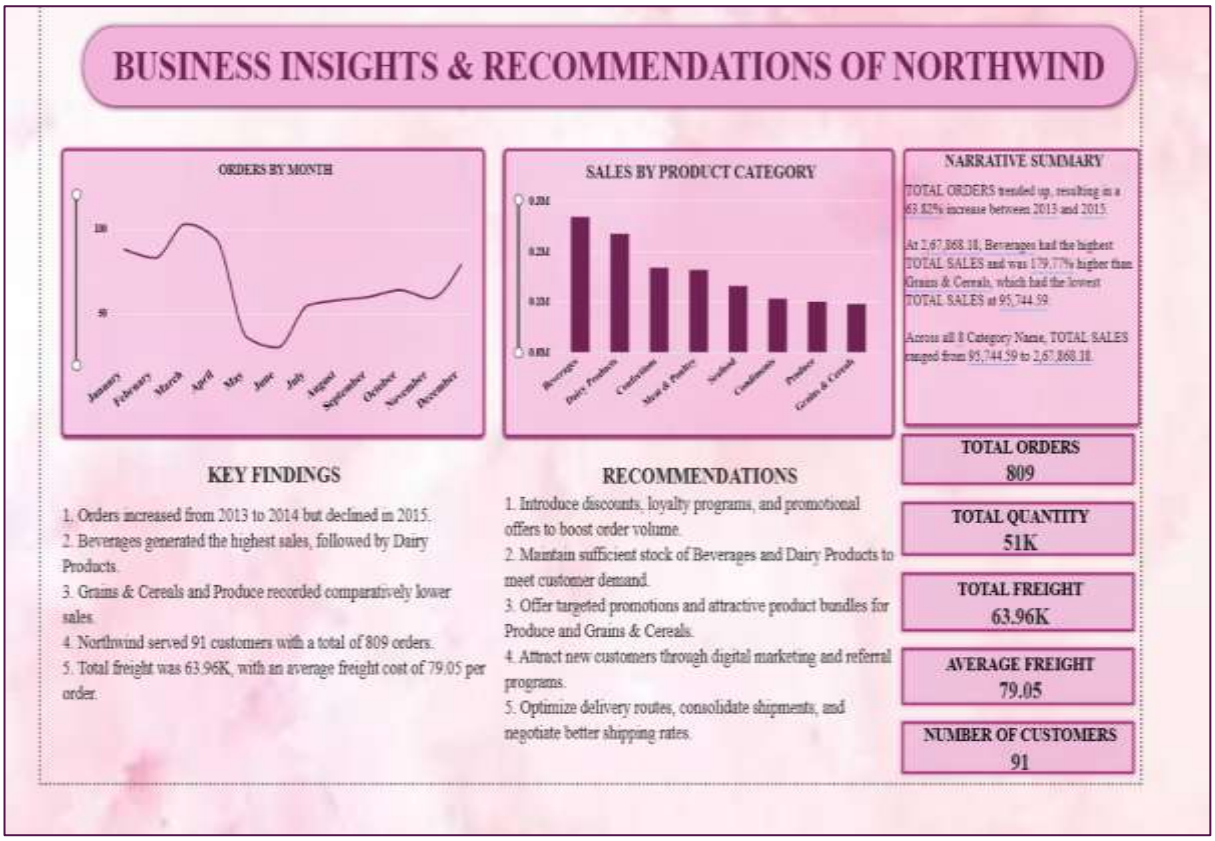
Key Insights:

- Margaret Peacock consistently achieved the highest number of orders across all three years (2013–2015).
- Employee performance varied yearly, highlighting the need to improve the productivity of lower-performing employees.

08.BUSINESS INSIGHTS:

- **Sales Performance:** Beverages is the highest-selling category, while Grains & Cereals has the lowest sales. The company should focus on promoting low-performing products.
- **Employee Performance:** Margaret Peacock achieved the highest performance with 151 orders, showing strong employee contribution. Employee training can help improve overall productivity.
- **Shipping & Operational Efficiency:** Total freight is 63.96K, with an average freight of 79.05. Monitoring shipping costs can help control expenses and improve operational efficiency.

Overall Insight: Northwind Traders can improve business performance by increasing product sales, maintaining customer relationships, improving employee productivity, and controlling shipping expenses.



09.BUSINESS RECOMMENDATIONS:

- **Increase Order Volume:** Introduce discounts, loyalty programs, and promotional offers to attract customers and encourage repeat purchases.
- **Maintain Product Availability:** Ensure sufficient stock of high-performing categories such as Beverages and Dairy Products to meet customer demand.
- **Improve Low-Performing Categories:** Offer targeted promotions and attractive product bundles for Produce and Grains & Cereals to increase their sales.
- **Expand Customer Base:** Attract new customers through digital marketing, referral programs, and customer engagement strategies to support business growth.
- **Optimize Shipping Costs:** Improve delivery routes, consolidate shipments, and negotiate better shipping rates to reduce freight expenses and improve operational efficiency.

Overall Recommendation: Focus on increasing sales, improving product demand, attracting new customers, and controlling operational costs to achieve sustainable business growth.

10.CONCLUSION:

The Northwind Traders Power BI project provides a clear understanding of the company's overall business performance through interactive dashboards and data analysis. The dashboard highlights sales performance, product categories, monthly order trends, employee performance, customer details, and shipping expenses. The analysis shows that Beverages is the highest-selling category, March recorded the highest number of orders (103), June recorded the lowest (30), and Margaret Peacock was the top-performing employee with 151 orders. Based on these findings, key recommendations include introducing promotional offers, maintaining sufficient product stock, improving low-performing categories, attracting new customers, and optimizing shipping costs. Power BI helped transform raw CSV data into meaningful visualisations using charts, KPI cards, and interactive filters, making it easier to identify trends, analyse business performance, and support effective decision-making.

Overall, this project enhanced practical knowledge of data analysis and demonstrated the importance of Power BI in improving business operations and strategic planning.

Northwind Traders Dataset Interactive Dashboard & Business Insights & Recommendation file link:
https://drive.google.com/file/d/1PulQCSm_PJwPfC_0k7bU-YiUJ8ophYHq/view?usp=drive_link